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BATTERY MODULE

Abstract

A battery module includes a plurality of battery cells with protruding terminal parts that include a first electrode terminal and a second electrode terminal. A frame part accommodates the plurality of battery cells, and busbars are electrically connected to the terminal parts of the plurality of battery cells. Each of the busbars is formed with insertion holes into which the terminal parts are inserted.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to and the benefit of Korean Patent Application No. 10-2024-0034111 filed in the Korean Intellectual Property Office on Mar. 11, 2024, the entire contents of which are incorporated herein by reference.

BACKGROUND

(a) Field

[0002] The present disclosure relates to a battery module capable of reducing heat generation by having a reduced height.

(b) Description of the Related Art

[0003] Secondary batteries are widely used in mobile devices, auxiliary power devices, or the like. In addition, secondary batteries are also attracting attention as a main power source for electric vehicles, hybrid electric vehicles, and plug-in hybrid electric vehicles, which have been proposed to solve various problems such as air pollution from conventional gasoline or diesel vehicles.

[0004] However, due to the need for high-output, large-capacity batteries in electric vehicles, battery modules are used in which multiple battery cells are stacked and then electrically connected in series and parallel. Each battery module includes a plurality of stacked battery cells, and terminals exposed at both ends of each battery cell are electrically connected to provide high voltage.

[0005] While the battery modules are being manufactured in a variety of methods and pack housing sizes to match shape and pack capacity, current packs have limitations in terms of scalability, serviceability, and recycling. Further, when using a plurality of conventional unit battery cells, the battery pack requires a structure capable of effectively dissipating heat generated during charging and discharging in order to provided high output.

SUMMARY OF THE DISCLOSURE

[0006] The present provides a battery module that is electrically connected with a part of an electrode terminal inserted into the inside of a busbar, and a portion where the electrode terminal is welded to the busbar is an edge part of the electrode terminal, thereby reducing heat generation.

[0007] The battery module according to an embodiment includes a plurality of battery cells that each include protruding terminal parts, with the terminal parts including a first electrode terminal and a second electrode terminal, a frame part accommodating the plurality of battery cells, and busbars electrically connected to the terminal parts of the plurality of battery cells, wherein an insertion hole is formed in each of the busbars, and the terminal parts are inserted into the insertion holes.

[0008] The insertion hole formed in each of the busbars includes a first insertion hole into which one of the first electrode terminals is inserted and electrically connected, and a second insertion hole into which one of the second electrode terminals is inserted and electrically connected.

[0009] Each of the busbars may include a first connection piece having the first insertion hole formed therein and through which the first electrode terminal is inserted and electrically connected, and a second connection piece extending at a side of the first connection piece, the second connection piece having the second insertion hole formed therein and through which the second electrode terminal of the battery cell is inserted and electrically connected.

[0010] A bending part may be formed between the first connection piece and the second connection piece in each of the busbars.

[0011] A through hole may be formed in the bending part.

[0012] A plurality of through hole may be formed in the bending part between the first connection piece and the second connection piece.

[0013] The plurality of through holes may include a first through hole formed in the bending part between the first connection piece and the second connection piece, and a second through hole formed in the bending part at a position spaced from the first through hole.

[0014] The first and second through holes may be formed such that lengths of the first and second through holes extend between the first connection piece and the second connection piece.
[0015] The first and second electrode terminals may be electrically connected by welding while inserted into the first and second insertion holes.
[0016] Inlet grooves may be formed on interior walls of the first insertion hole and the second insertion hole.
[0017] Insertion protrusions inserted into the inlet groove may protrude from side surfaces of the first electrode terminal and the second electrode terminal.
[0018] According to another embodiment, a battery module may include a first battery cell including an electrode terminal; a second battery cell including an electrode terminal; and a busbar with first and second insertion holes formed therein, with the electrode terminal of the first battery cell being positioned in the first insertion hole of the busbar and the terminal part of the of the second battery cell being positioned in the second insertion hole of the busbar.
[0019] An edge part of the electrode terminal of the first battery cell may be welded to the busbar, and an edge part of the electrode terminal of the second battery cell is welded to the busbar.
[0020] The first insertion hole may be formed in a first connection piece of the busbar, and the second insertion hole may be formed in a second connection piece of the busbar, and the first connection piece and the second connection piece may be connected by a bendable part of the busbar.
[0021] An edge part of the electrode terminal of the first battery cell may be welded to the first connection piece of the busbar, and an edge part of the electrode terminal of the second battery cell may be welded to the second connection piece of the busbar.
[0022] According to a further embodiment, a battery module may include a plurality of battery cells, each of the battery cells including first and second electrode terminals; a plurality of busbars, each of the plurality of busbars having first and second insertion holes formed therein, with the first electrode terminal of each of the battery cells being electrically connected to one of the busbars with the first electrode terminal being inserted into the first insertion hole of one of the busbars, and the second electrode terminal of each of the battery cells being electrically connected to another of the busbars with the second electrode terminal being inserted into the second insertion hole of the another of the plurality of bus cars.
[0023] An edge part of each of the first electrode terminals may be welded to the busbar having the first insertion hole that receives the first electrode terminal, and an edge part of each of the second electrode terminals may be welded to the busbar having the second insertion hole that received the second electrode terminal.
[0024] Each of the first insertion holes may be formed in a first connection piece of the busbar, and each of the second insertion holes may be formed in a second connection piece of the busbar, and the first connection piece and the second connection piece are connected by a bendable part of the busbar.
[0025] An edge part of each of the first electrode terminals may welded to the first connection piece of the busbar having the first insertion hole that receives the first electrode terminal, and an edge part of each of the second electrode terminals may be welded to the second connection piece of the busbar having the second insertion hole that receives the second electrode terminal.
[0026] According to an embodiment of the present disclosure, a part of the electrode terminal is electrically connected by welding while it is inserted into an insertion hole formed in the busbar. Accordingly, the part where the electrode terminal and the busbar are welded is an edge part of the electrode terminal, thereby allowing current to flow along the edge part of the electrode terminal, and, thus, reducing heat generation in the busbar.
[0027] According to an embodiment of the present disclosure, a part of the electrode terminal is electrically connected by welding while it is inserted into an insertion hole formed in the busbar,

and, thus, the overall height of the battery module is reduced, thereby increasing the cell capacity because of an increase in the amount of power per unit volume.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0028] FIG. **1** is a perspective view schematically illustrating a battery module according to an embodiment of the present disclosure.

[0029] FIG. **2** is a perspective view of the main part schematically illustrating a state in which a busbar is connected to the battery cell of FIG. **1**.

[0030] FIG. **3** is a perspective view of the main part schematically illustrating a state in which the electrode terminal of FIG. **1** is inserted into the busbar and connected.

[0031] FIG. **4** is a perspective view schematically illustrating a busbar according to an embodiment of the present disclosure.

[0032] FIG. **5** is a side view schematically illustrating the busbar of FIG. **4**.

[0033] FIG. **6** is a side view of the main part schematically illustrating a state in which an electrode terminal is connected to the busbar of FIG. **4**.

[0034] FIG. **7** is a perspective view of the main part schematically illustrating a state in which an electrode terminal of a battery module according to a second embodiment of the present disclosure is inserted into the busbar and connected.

[0035] FIG. **8** is a perspective view schematically illustrating the busbar of FIG. **7**.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0036] The present disclosure will be described more fully hereinafter with reference to the accompanying drawings, in which exemplary embodiments of the disclosure are shown. As those skilled in the art would realize, the described embodiments may be modified in various different ways without departing from the spirit or scope of the present disclosure. The drawings and description are to be regarded as illustrative in nature and not restrictive. Like reference numerals designate like elements throughout the specification.

[0037] FIG. **1** is a perspective view schematically illustrating a battery module according to an embodiment of the present disclosure, and FIG. **2** is a perspective view of the main part schematically illustrating a state in which a busbar is connected to the battery cell of FIG. **1**.

[0038] As shown in FIGS. **1** and **2**, a battery module **100** according to an embodiment of the present disclosure includes a plurality of battery cells **10** with protruding terminal parts **11** including a first electrode terminal **11***a* and a second electrode terminal **11***b*, a frame part **20** accommodating the plurality of battery cells **10**, and busbars **30** electrically connected to the terminal parts **11** of the plurality of battery cells **10**. The busbars **30** are formed with insertion holes **21***a* and **23***a* into which the terminal parts **11** are inserted.

[0039] The frame part **20** may include a first side frame **21** supporting one side surface of the plurality of battery cells **10**, a second side frame **22** supporting the other side surface of the plurality of battery cells **10**, a first end frame **23** with ends that are connected to ends of the first side frame **21** and the second side frame **22**, and a second end frame **24** with ends that are connected to each end of the first side frame **21** and the second side frame **22**.

[0040] The first side frame **21** may be installed to support one side of the plurality of battery cells **10**. The second side frame **22** may be installed to support the other side of the plurality of battery cells **10**. The first end frame **23** may be installed to connect one end of each of the first side frame **21** and the second side frame **22**. That is, one end of the first end frame **23** may be connected to one end of the first side frame **21** by a fastening member, and the other end of the first end frame **23** may be connected to one end of the second side frame **22** by the fastening member.

[0041] The second end frame **24** may be installed to connect the other ends of the first side frame

21 and the second side frame **22**. That is, one end of the second end frame **24** may be connected to the other end of the first side frame **21** by the fastening member, and the other end of the second end member **24** may be connected to the other end of the second side frame **22** by the fastening member.

[0042] The battery cells **10** may be inserted into the frame part **20** in plural numbers arranged adjacent to each other in the first direction (x-axis direction). Each of the battery cells **10** has the terminal part **11** including a first electrode terminal **11***a* and a second electrode terminal **11***b* projecting upwardly. The terminal parts **11** of neighboring battery cells **10** may be electrically connected to each other by the busbars **30**. Each terminal part **11** may be electrically connected to one of the busbars **30** with a part inserted into the busbar **30**. As such, a part of the terminal part **11** is inserted into the busbar **30** so that a part where the terminal part **11** and the busbar **30** are welded is changed to an edge part of the terminal part, allowing current to flow along the edge part of the electrode terminal, thereby reducing heat generation in the busbar **30**.

[0043] In addition, a part of the terminal part **11** is inserted into the busbar **30** to reduce the overall height of the battery module **100**, thereby enabling the implementation of the battery module **100** with a relatively low height. This will be described in detail below while describing the busbar **30**.

[0044] Each busbar **30** electrically connects the plurality of battery cells **10** to each other and may include a first insertion hole **31***a* into which the first electrode terminal **11***a* is inserted and electrically connected, and a second insertion hole **33***a* into which the second electrode terminal **11***b* is inserted and electrically connected.

[0045] FIG. **3** is a perspective view of the main part schematically illustrating a state in which the electrode terminal of FIG. **1** is inserted into the busbar and connected, FIG. **4** is a perspective view schematically illustrating a busbar according to an embodiment of the present disclosure, and FIG. **5** is a side view schematically illustrating the busbar of FIG. **4**.

[0046] Referring to FIGS. **3** to **5**, each busbar **30** may include a first connection piece **31** having the first insertion hole **31***a* through which a first electrode terminal **11***a* of a battery cell **10** is inserted and electrically connected. Each busbar **30** may further include a second connection piece **33** extending to a side surface of the first connection piece **31** and having the second insertion hole **33***a* through which a second electrode terminal **11***b* of a battery cell **10** is inserted and electrically connected.

[0047] The first connection piece **31** and the second connection piece **33** are connected to electrically connect a pair of adjacent battery cells **10** to each other, and the connection pieces **31** and **33** may be formed in the same or similar shapes and connected to each other.

[0048] The first connection piece **31** may have a substantially rectangular or square shape, and the first insertion hole **31***a* through which the first electrode terminal **11***a* is inserted and electrically connected is formed in the first connection piece **31**. The first insertion hole **31***a* may be formed through the first connection piece **31** in a shape corresponding to the shape of the first electrode terminal **11***a*. That is, the first insertion hole **31***a* may be formed in a rectangular shape corresponding to the shape of the first electrode terminal **11***a*, which is rectangular in the depicted embodiment. If the first electrode terminal **11***a* is changed to a round shape, the first insertion hole **31***a* may also be changed to a corresponding round shape.

[0049] The first electrode terminal **11***a* may be electrically connected to the bust bar **30** by welding while insert in the first insertion hole **31***a*. That is, the first electrode terminal **11***a* may be electrically connected to the first connection piece **31** by butt welding while inserted into the first insertion hole **31***a*.

[0050] The second connection piece **33** is connected to the side surface of the first connection piece **31** and may be electrically connected to the second electrode terminal **11***b* of the adjacent battery cell **10**. The second connection piece **33** may have a substantially rectangular or square shape, and the second insertion hole **33***a* may be formed therein into which the second electrode terminal **11***b* is inserted and electrically connected. The second insertion hole **33***a* may be formed in a shape corresponding to the shape of the second electrode terminal **11**b. That is, the second insertion hole **33**a may be formed through the second electrode terminal **11**b in a rectangular shape corresponding to the shape of the second electrode terminal **11**b, which is rectangular in shape in the depicted embodiment. If the second electrode terminal **11**a is changed to a round shape, the first insertion hole **31**a may also be changed to a corresponding round shape.

[0051] The second electrode terminal **11**b may be electrically connected by welding while inserted in the second insertion hole **33**a. That is, the second electrode terminal **11**b may be electrically connected to the second connection piece **33** by butt welding while inserted into the second insertion hole **33**a.

[0052] Thus, the first and second connection pieces **31** and **33** and the first and second electrode terminals **11**a and **11**b are connected by welding, and, in particular, may be electrically connected to each other by butt welding in the present embodiment.

The first connection piece **31** and the second connection piece **33** may be connected by a bending part **35**. One side of the bending part **35** may be connected to the edge of the first connection piece **31**, and the other side of the bending part **35** may be connected to one side of the second connection piece **33**. The central part of the bending part **35** may be bent into a round shape toward the battery cell **10**.

[0053] The bending part **35** has an area smaller than the area of the first connection piece **31** or the area of second connection piece **33**. The bending part **35** may connect the first connection piece **31** and the second connection piece **33** in a bent state. As such, the formation of the bending part **35** between the first connection piece **31** and the second connection piece **33** allows for elastic deformation by the bending shape during the process of connecting the first and second connection pieces **31** and **33** to each of the first and second electrode terminals **11**a and **11**b. Thus, a smoother connection is possible.

[0054] It is also possible for one or more through holes **36** to be formed in the bending part **35**. In particular, a plurality of through holes **36** may be formed in the bending part **35**. The through holes **36** allow for smoother elastic deformation of the bending part **35** during the process of connecting the busbar **30** to the first and second electrode terminals **11**a and **11**b. The through hole **36** may include a first through hole **36**a formed in the bending part **35** between the first connection piece **31** and the second connection piece **33**, and a second through hole **36**b formed between the first connection piece **31** and the second connection piece **33** in the bending part **35** at a position spaced apart from the first through hole **36**a.

[0055] The first through hole **36**a may be formed through the bending part **35** with a long length toward the first connection piece **31** and the second connection piece **33**.

[0056] The first through hole **36**a may be formed in a long hole shape between the first connection piece **31** and the second connection piece **33**. The first through hole **36**a thereby allows for easier elastic deformation of the bending part **35** in the process of connecting the first connection piece **31** and the second connection piece **33** to the first electrode terminal **11**a and the second electrode terminal **11**b, respectively. Thus, the first through hole **36**a improves the flexibility of the connection between the busbar **30** to the terminal part **11**.

[0057] The second through hole **36**b has the same or similar shape as the first through hole **36**a, and the second through hole **36**b may be formed in the bending part **35** at a position spaced apart from the position where the first through hole **36**a is formed. The second through hole **36**b may be formed through the bending part **35** with a long length toward the first connection piece **31** and the second connection piece **33**. The second through hole **36**b may be formed in a long hole shape between the first connection piece **31** and the second connection piece **33**. The second through hole **36**b therefore has a similar effect to that of the first through hole **36**a in allowing for easier elastic deformation of the bending part **35** in the process of connecting the first connection piece **31** and the second connection piece **33** to the first electrode terminal **11**a and the second electrode terminal **11**b, respectively. Thus, the second through hole **36**b improve the flexibility of the connection between the busbar **30** to the terminal part **11**.

[0058] The first and second through holes **36***a* and **36***b* are as having the same or similar shapes, but the disclosure not necessarily limited thereto, as the through holes **36***a* and **36***b* may have different shapes.

[0059] As described above, the battery module **100** of the present embodiment may be electrically connected by welding, with parts of the first and second electrode terminals **11***a* and **11***b* inserted into the first and second insertion holes **31***a* and **33***a* formed in each of the first and second connection pieces **31** and **33** of the busbar **30**. Accordingly, the part where the terminal part **110** and the busbar are welded is an edge portion of the terminal part, allowing current to flow to the busbar **30** along the edge part of the electrode terminal, thereby reducing heat generation in the busbar **30**. In addition, the height from the bottom of the battery cell **10** of the battery module **100** to the top of the busbar **30** is reduced. Further, the amount of power per unit volume of the battery module **100** may be increased, thereby increasing the capacity of the battery cell in the battery module.

[0060] FIG. **7** is a perspective view schematically illustrating a state in which the electrode terminal of the battery module according to the second embodiment of the present disclosure is inserted into the busbar and connected, and FIG. **8** is a perspective view schematically illustrating the busbar of FIG. **7**. The same reference numbers as in FIGS. **1** to **6** refer to the same or similar members with the same or similar functions. Thus, detailed descriptions of the same reference numbers will be omitted.

[0061] As shown in FIGS. **7** and **8**, the busbar **30** of the battery module **200** according to the second embodiment of the present disclosure may include the first connection piece **31** having the first insertion hole **31***a* through which the first electrode terminal **11***a* of the battery cell **10** is inserted and electrically connected. The busbar **30** may also include the second connection piece **33** extending to the side of the first connection piece **31** and having the second insertion hole **33***a* through which the second electrode terminal **11***b* of the battery cell **10** is inserted and electrically connected.

[0062] Inlet grooves **37** may be formed on interior walls of the first insertion hole **31***a* and the second insertion hole **33***a.* A plurality of the inlet grooves **37** may be formed in each of the first insertion hole **31***a* and the second insertion hole **33***a.* Insertion protrusions **11***c* protruding from side surfaces of the first electrode terminal **11***a* and the second electrode terminal **11***b* may be inserted into the inlet grooves **37**. In particular, one of the insertion protrusions **11***c* is inserted into the inlet groove **37** formed in each of the first and second insertion holes **31***a* and **33***a,* and the insertion protrusions **11***c* may be inserted in a press-fitted state into the first and second insertion holes **31***a* and **33***a.* Thus, the busbar **30** may be more stably fixed to the first and second electrode terminals **11***a* and **11***b.*

[0063] While this disclosure has been described in connection with what is presently considered to be practical embodiments, it is to be understood that the disclosure is not limited to the disclosed embodiments. On the contrary, the disclosure covers various modifications and equivalent arrangements.

DESCRIPTION OF SYMBOLS

[0064] **10** . . . Battery cell **11** . . . Terminal part [0065] **11***a* . . . First electrode terminal **11***b* . . . Second electrode terminal [0066] **20** . . . Frame part **21** . . . First side frame [0067] **22** . . . Second side frame **23** . . . First end frame [0068] **24** . . . Second end frame **30** . . . Busbar [0069] **31** . . . First connection piece **31***a* . . . First insertion hole [0070] **33** . . . Second connection piece **33***a* . . . Second insertion hole [0071] **35** . . . Bending part **36** . . . Through hole [0072] **36***a* . . . First through hole **36***b* . . . Second through hole

Claims

1. A battery module comprising: a plurality of battery cells that each include protruding terminal parts, the terminal parts each include a first electrode terminal and a second electrode terminal; a frame part accommodating the plurality of battery cells; and busbars electrically connected to the terminal parts of the plurality of battery cells, wherein an insertion hole is formed in each of the busbars, and the terminal parts are inserted into the insertion holes.
2. The battery module as claimed in claim 1, wherein the insertion hole formed in each of the busbars includes a first insertion hole into which one of the first electrode terminals is inserted and electrically connected, and a second insertion hole into which one of the second electrode terminals is inserted and electrically connected.
3. The battery module as claimed in claim 2, wherein each of the busbars comprises: a first connection piece having the first insertion hole formed therein and through which the first electrode terminals is inserted and electrically connected; and a second connection piece extending at a side of the first connection piece, the second connection piece having the second insertion hole formed therein and through which the second electrode terminal of the battery cell is inserted and electrically connected.
4. The battery module as claimed in claim 3, wherein a bending part is formed between the first connection piece and the second connection piece in each of the busbars.
5. The battery module as claimed in claim 4, wherein a through hole is formed in the bending part.
6. The battery module as claimed in claim 4, wherein a plurality of through holes is formed in the bending part between the first connection piece and the second connection piece.
7. The battery module as claimed in claim 6, wherein the plurality of through holes comprises: a first through hole formed in the bending part between the first connection piece and the second connection piece; and a second through hole formed in the bending part at a position spaced from the first through hole.
8. The battery module as claimed in claim 7, wherein the first and second through holes are formed such that a length of the first and second through holes extends between the first connection piece and the second connection piece.
9. The battery module as claimed in claim 2, wherein the first and second electrode terminals are electrically connected by welding while being inserted into the first and second insertion holes.
10. The battery module as claimed in claim 3, wherein inlet grooves are formed on interior walls of the first insertion hole and the second insertion hole.
11. The battery module as claimed in claim 10, wherein insertion protrusions protrude from side surfaces of the first electrode terminal and the second electrode terminal, and each of the protrusions is inserted into one of the inlet grooves.
12. A battery module comprising: a first battery cell including an electrode terminal; a second battery cell including an electrode terminal; and a busbar with first and second insertion holes formed therein, wherein the electrode terminal of the first battery cell is positioned in the first insertion hole of the busbar and the terminal part of the of the second battery cell is positioned in the second insertion hole of the busbar.
13. A battery module as claimed in claim 12, wherein an edge part of the electrode terminal of the first battery cell is welded to the busbar, and an edge part of the electrode terminal of the second battery cell is welded to the busbar.
14. A battery module as claimed in claim 12, wherein the first insertion hole is formed in a first connection piece of the busbar, and the second insertion hole is formed in a second connection piece of the busbar, and wherein the first connection piece and the second connection piece are connected by a bendable part of the busbar.
15. A battery module as claimed in claim 14, wherein an edge part of the electrode terminal of the first battery cell is welded to the first connection piece of the busbar, and an edge part of the electrode terminal of the second battery cell is welded to the second connection piece of the busbar.

16. A battery module comprising: a plurality of battery cells, each of the battery cells including first and second electrode terminals; a plurality of busbars, each of the plurality of busbars having first and second insertion holes formed therein, wherein the first electrode terminal of each of the battery cells is electrically connected to one of the busbars with the first electrode terminal being inserted into the first insertion hole of one of the busbars, and wherein the second electrode terminal of each of the battery cells is electrically connected to another of the busbars with the second electrode terminal being inserted into the second insertion hole of the another of the plurality of bus cars.

17. A battery module as claimed in claim 16, wherein an edge part of each of the first electrode terminals is welded to the busbar having the first insertion hole that receives the first electrode terminal, and an edge part of each of the second electrode terminals is welded to the busbar having the second insertion hole that received the second electrode terminal.

18. A battery module as claimed in claim 16, wherein each of the first insertion holes is formed in a first connection piece of the busbar, and each of the second insertion holes is formed in a second connection piece of the busbar, and wherein the first connection piece and the second connection piece are connected by a bendable part of the busbar.

19. A battery module as claimed in claim 18, wherein an edge part of each of the first electrode terminals is welded to the first connection piece of the busbar having the first insertion hole that receives the first electrode terminal, and an edge part of each of the second electrode terminals is welded to the second connection piece of the busbar having the second insertion hole that receives the second electrode terminal.